

 VIT-AP UNIVERSITY	Continuous Assessment Test – Winter Sem(2024-25) -JAN 2025	
	Maximum Marks: 50	Duration: 90 Mins
Course Code: CSE1008	Course Title: Theory of Computation	
Set No: I	Exam Type : Closed Book	School: SCOPE
Date: 30-01-25	Slot: DI+TDI+TDDI	Session: FN
Keeping mobile phone/smart watch, even in ‘off’ position is treated as exam malpractice		
General Instructions if any Open Book/Open Notebook/Closed Book:		
1. “fx series” - non Programmable calculator are permitted : NO		
2. Reference tables permitted : NO		

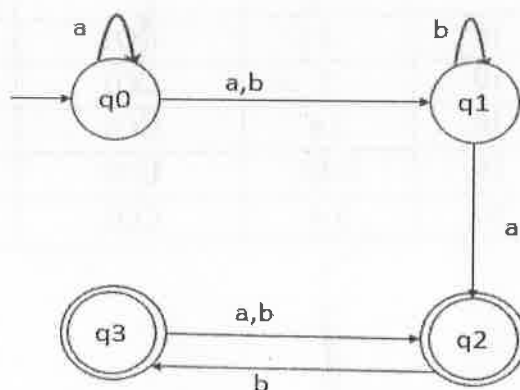
PART – A: Answer any ALL Questions, Each Question Carries 10 Marks (5×10=50 Marks)

1. Explain the significance of multiple paths in an NFA. Can a string be accepted if at least one path leads to an accept state? Justify your answer. Construct a DFA for the language accepting strings starting with ‘101’.

(10M)

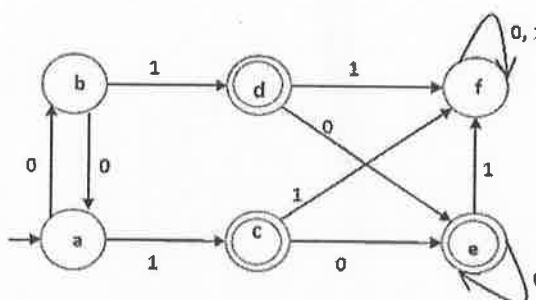
2. What is the difference between DFA and NFA? Consider the following Non-deterministic finite automata (NFA) and convert that NFA into equivalent Deterministic Finite Automata (DFA).

(10M)

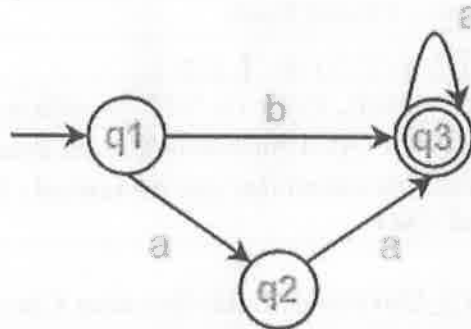


3. Minimize the given Deterministic Finite Automaton (DFA) and provide a detailed explanation of each step involved.

(10M)



4. A) Let $\Sigma = \{a, b\}$. Write a regular expression to define a language consisting of strings w such that w is of even length. (5 M)
 B) Convert the following RE into its equivalent FA: $10+(0+11)0^*1$ (5 M)
5. Construct a regular expression corresponding to the automata given below. (10 M)



QP MAPPING

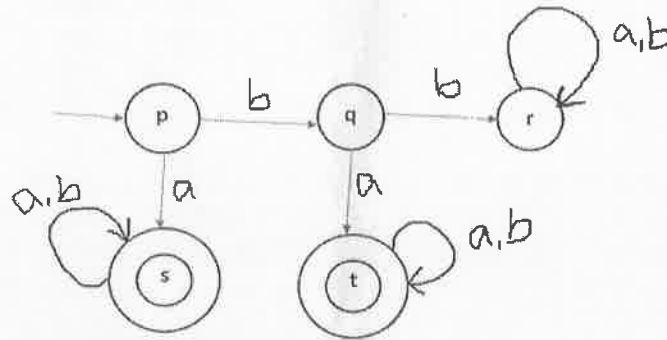
Q. No.	E/A/T	Module Number	Marks	BL	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped
Q1	E	1	10	2	CO1	PO1		
Q2	E	1	10	2	CO1	PO3		
Q3	T	1	10	3,4	CO1	PO3		
Q4	A	2	10	3,4,2	CO2	PO2		
Q5	A	2	10	3,4	CO2	PO1		

 VIT-AP UNIVERSITY	Continuous Assessment Test – Winter Sem(2024-25) -JAN 2025	
Course Code: CSE1008	Maximum Marks: 50	Duration: 90 Mins
Set No: 2	Course Title: Theory of Computation	
Date: 28-01-25	Exam Type : Closed Book	School: SCOPE
	Slot: B2 + TB2 + TB32	Session: AN
Keeping mobile phone/smart watch, even in 'off' position is treated as exam malpractice		
General Instructions if any Open Book/Open Notebook/Closed Book:		
1. "fx series" - non Programmable calculator are permitted : NO 2. Reference tables permitted : NO		

PART – A: Answer any ALL Questions, Each Question Carries 10 Marks (5×10=50 Marks)

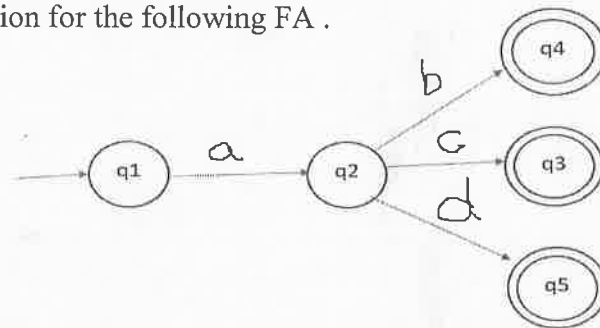
1. Design non-deterministic finite automata which accept 00 or 11 at the end of a string. String must contain 0, 1 in it, e.g., 01010100 but not 0001110101. **(10 M)**

2. A deterministic finite automation (DFA) with alphabet $\Sigma = \{a, b\}$ is given below. Find the minimal DFA **(10 M)**



3. Construct a DFA over $\{0,1\}$ that accepts the following.
A) Binary strings such that the third symbol from the right end is 1. **(5 M)**
B) Set of all strings with three consecutive 0's **(5 M)**

4. Find regular expression for the following FA . **(10 M)**



5. Given Regular language

$L1 = ((a+b)(a+b)(a+b))^*$. Find the equivalent DFA. **(10 M)**

 VIT-AP UNIVERSITY	Continuous Assessment Test – Winter Sem (2024-25) -JAN 2025	
	Maximum Marks: 50	Duration: 90 Mins
Course Code: CSE1008	Course Title: Theory of Computation	
Set No: 3	Exam Type : Closed Book	School: SCOPE
Date: 28-01-25	Slot: B1 + TB1 + TB B1	Session: FN
Keeping mobile phone/smart watch, even in 'off' position is treated as exam malpractice		
General Instructions if any Open Book/Open Notebook/Closed Book:		
1. "fx series" - non Programmable calculator are permitted : NO		
2. Reference tables permitted : NO		

PART – A: Answer any ALL Questions, Each Question Carries 10 Marks (5×10=50 Marks)

1. Design a DFA for the language $L = \{w \mid \text{where } |w| \text{ is multiples of 3 and string starting with 'a'}\}$, $\Sigma = \{a, b\}$. Perform extended transition function for the following strings. (10 Marks)

- i) abb
ii) abab

2. Consider the following ϵ -NFA and convert into DFA.

(10 Marks)

Δ	ϵ	a	b	C
$\rightarrow p$	$\{q, r\}$	$\emptyset$	$\{q\}$	$\{r\}$
Q	$\{p\}$	$\{p\}$	$\{r\}$	$\{p, q\}$
$*r$	$\emptyset$	$\emptyset$	$\emptyset$	Φ

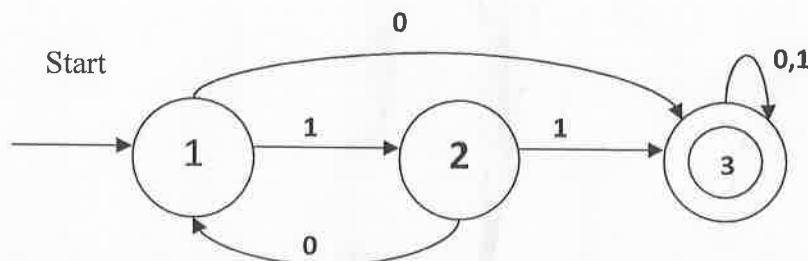
3. Consider the following DFA. Find the minimized version of the DFA.

(10 Marks)

δ	0	1
$\rightarrow a$	b	a
b	a	c
$*c$	d	b
$*d$	d	a
$*e$	d	F
f	g	E
g	f	G
h	g	D

4. Consider the following DFA and find the corresponding regular expression.

(10 Marks)



5. a) Find the regular expression for the language $L = \{\text{Set of odd number of 'a' followed by any number of b's followed by any number of c's}\}$ (3 Marks)
- b) Convert the following Regular expression into FA. (7 Marks)
- $R.E = (0+1)0^*101^*$

QP MAPPING

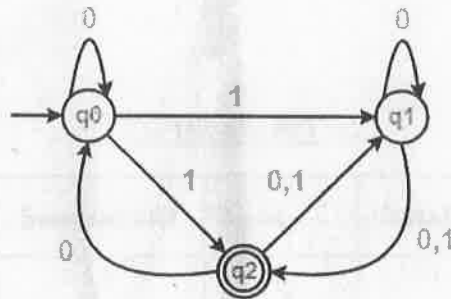
Q. No.	E/A/T	Module Number	Marks	BL	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped
Q1	T	1	10	3	CO1	PO1, PO3	-	-
Q2	A	1	10	3	CO1	PO1, PO3	-	-
Q3	A	1	10	3	CO1	PO1, PO3	-	-
Q4	A	2	10	3	CO2	PO1, PO2, PO3	-	-
Q5	E	2	10	2	CO2	PO1, PO2, PO3	-	-

 VIT-AP UNIVERSITY	Continuous Assessment Test – Winter Sem(2024-25) -JAN 2025	
Course Code: CSE1008	Maximum Marks: 50	Duration: 90 Mins
Set No: 6	Course Title: Theory of Computation	School: SCOPE
Date: 30-01-25	Exam Type : Closed Book	Session: AN
Slot: D2 + TD2 + TDD2		
Keeping mobile phone/smart watch, even in 'off' position is treated as exam malpractice		
General Instructions if any Closed Book:		
1. "fx series" - non Programmable calculator are permitted : NO 2. Reference tables permitted : NO		

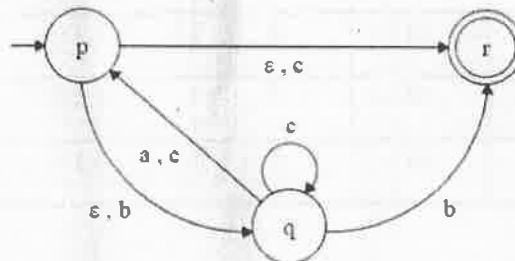
PART – A: Answer any ALL Questions, Each Question Carries 10 Marks (5×10=50 Marks)

1. Design a NFA for the language of all strings over $\{0, 1\}$ that does not end with one of 100, and 11. (10 M)

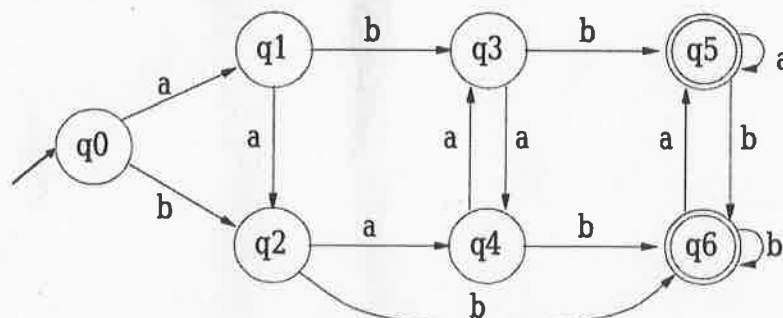
2. (a) Convert the following NFA to DFA. (5 M)



- (b) Convert the following epsilon-NFA to NFA. (5 M)

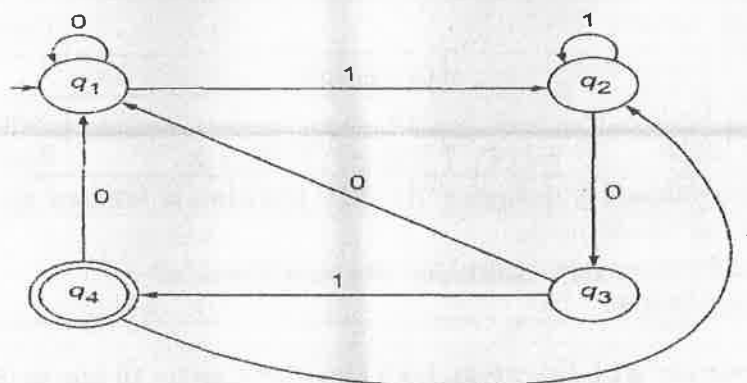


3. Minimize the state of the following Deterministic Finite Automata. (10 M)



4. Convert the following Finite Automata to Regular Expression.

(10 M)



5. Construct Finite Automata for the Regular Expression, $R = (aba + bb)^* a + ab$.

(10 M)

QP MAPPING

Q. No.	E/A/T	Module Number	Marks	BL	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped
Q1	A	1	10	5	1,2	1,2,3,4	1,2,3,4	1,2,3
Q2	T	1	10	3	1,2	1,2,3,4	1,2,3,4	1,2,3
Q3	A	1	10	3	1,2	1,2,3,4	1,2,3,4	1,2,3
Q4	A	2	10	3	1,2	1,2,3,4	1,2,3,4	1,2,3
Q5	T	2	10	5	1,2	1,2,3,4	1,2,3,4	1,2,3